Settings of test systems

I. MODIFIED IEEE 9-BUS SYSTEM

A. Connection structure

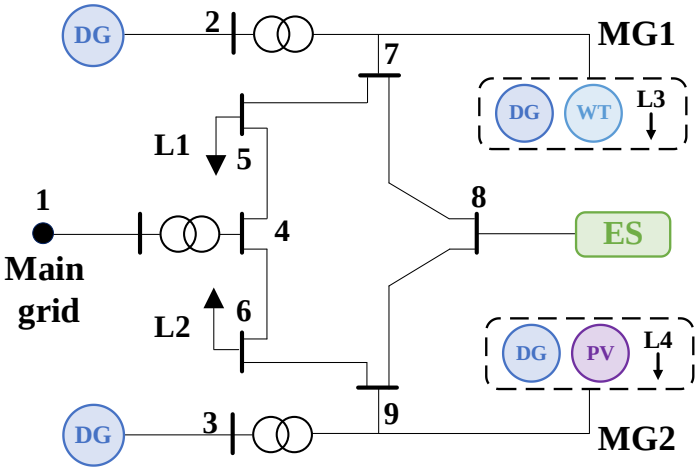


Fig. 1. The diagram of the modified IEEE 9-bus system.

B. Network parameters

TABLE I
NETWORK PARAMETERS

Line index	From bus	To bus	r (Ω)	x (Ω)
1	1	4	0	0.0576
2	4	6	0.017	0.092
3	6	9	0.039	0.17
4	3	9	0	0.0586
5	8	9	0.0119	0.1008
6	7	8	0.0085	0.072
7	2	7	0	0.0625
8	5	7	0.032	0.161
9	4	5	0.01	0.085

C. Units parameters

TABLE II
ENERGY STORAGE PARAMETERS

E	$P_i^{E,max}$ (kW)	$E_{i,max}$ (kWh)	$E_{i,min}$ (kWh)	b_i^E ($/kWh)	c_i^E ($)	η_i
8	100	800	0	1.351e-6	0.074	0.95

TABLE III
GENERATORS PARAMETERS

G	$P_i^{G,max}$ (kW)	$P_i^{G,min}$ (kW)	R_i^G (kW/h)	a_i^G ($/kW²h)	b_i^G ($/kWh)	c_i^G ($)
2	900	0	500	4.05e-5	0.028	0.3784

G	$P_i^{G,max}$ (kW)	$P_i^{G,min}$ (kW)	R_i^G (kW/h)	a_i^G (\$/kW²h)	b_i^G (\$/kWh)	c_i^G (\$)
3	1200	0	800	3.53e-5	0.032	0.7432
7	600	0	400	3.00e-5	0.025	0.4124
9	1000	0	500	3.20e-5	0.021	0.3786

II. MODIFIED IEEE 33-BUS SYSTEM

A. Connection structure

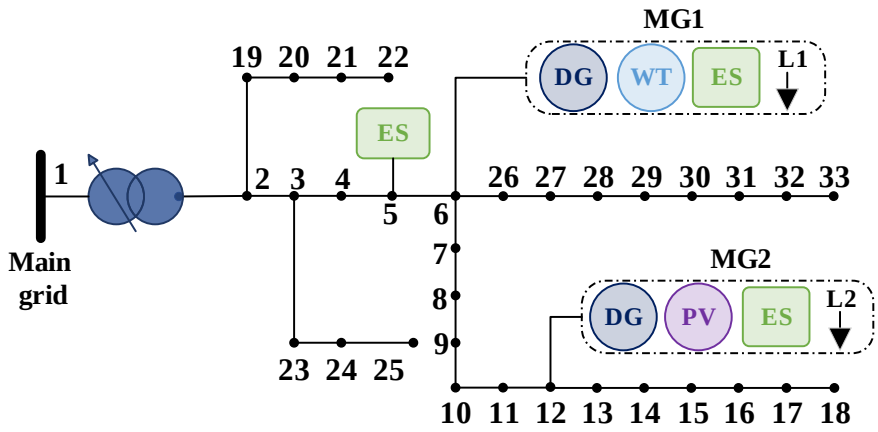


Fig. 2. The diagram of the modified IEEE 33-bus system.

B. Network parameters

TABLE IV
NETWORK PARAMETERS

Line index	From bus	To bus	r (Ω)	x (Ω)
1	1	2	0.0922	0.0470
2	2	3	0.4390	0.2511
3	3	4	0.3660	0.1864
4	4	5	0.3811	0.1941
5	5	6	0.8190	0.7070
6	6	7	0.1872	0.6188
7	7	8	0.7114	0.2351
8	8	9	1.0300	0.7400
9	9	10	1.0440	0.7400
10	10	11	0.1966	0.0650
11	11	12	0.3744	0.1238
12	12	13	1.4680	1.1550
13	13	14	0.5416	0.7129
14	14	15	0.5910	0.5260
15	15	16	0.7463	0.5450
16	16	17	1.2890	1.7210
17	17	18	0.7320	0.5740

Line index	From bus	To bus	r (Ω)	x (Ω)
18	2	19	0.1640	0.1565
19	19	20	1.5042	1.3554
20	20	21	0.4095	0.4784
21	21	22	0.7089	0.9373
22	3	23	0.4512	0.3083
23	23	24	0.8980	0.7091
24	24	25	0.8960	0.7011
25	6	26	0.2030	0.1034
26	26	27	0.2842	0.1447
27	27	28	1.0590	0.9337
28	28	29	0.8042	0.7006
29	29	30	0.5075	0.2585
30	30	31	0.9744	0.9630
31	31	32	0.3105	0.3619
32	32	33	0.3410	0.5302

C. Units parameters

TABLE V
ENERGY STORAGE PARAMETERS

E	$P_i^{E,max}$ (kW)	$E_{i,max}$ (kWh)	$E_{i,min}$ (kWh)	b_i^E ($/kWh)	c_i^E ($)	η_i
5	500	1000	0	1.351e-6	0.074	0.95
6	100	300	0	2.045e-6	0.085	0.95
12	100	200	0	1.854e-6	0.069	0.95

TABLE VI
GENERATOR PARAMETERS

G	$P_i^{G,max}$ (kW)	$P_i^{G,min}$ (kW)	R_i^G (kW/h)	a_i^G ($/kW²h)	b_i^G ($/kWh)	c_i^G ($)
6	1000	0	500	3.00e-5	0.029	0.3784
10	1000	0	500	4.50e-5	0.042	0.7462
12	900	0	400	3.20e-5	0.024	0.4124
19	1200	0	600	3.40e-5	0.052	0.3898
23	1500	0	800	3.90e-5	0.032	0.6582
30	1000	0	500	4.10e-5	0.028	0.4686

III. A REAL 265-BUS SYSTEM

A. Connection structure

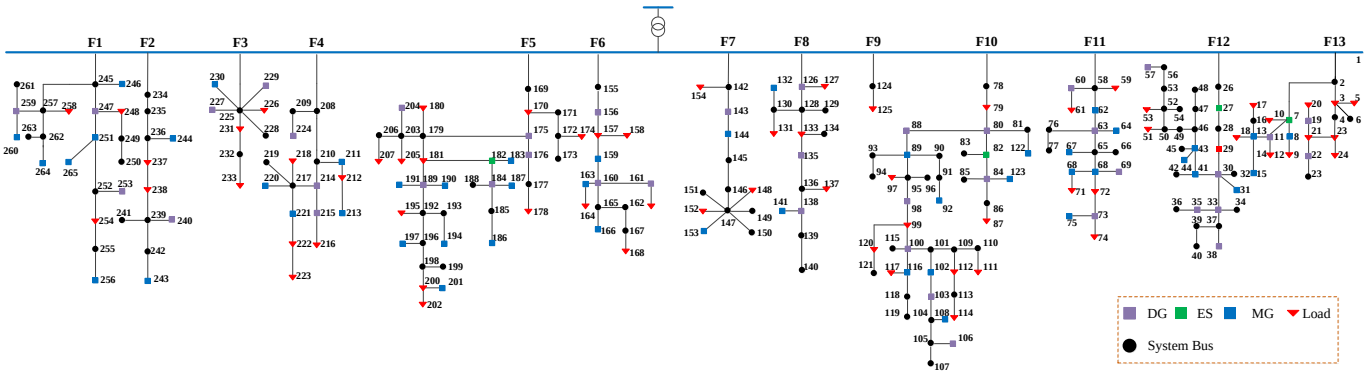


Fig. 3. The diagram of the real 265-bus system.

B. Network parameters

TABLE VII
GENERATOR PARAMETERS

Line index	From bus	To bus	r (Ω)	x (Ω)	Line index	From bus	To bus	r (Ω)	x (Ω)
1	1	2	0.00122	0.00202	133	126	127	0.02133	0.03543
2	1	26	0.00112	0.00186	134	126	128	0.00152	0.00253
3	1	58	0.09001	0.14947	135	128	129	0.00244	0.00405
4	1	78	0.00157	0.00261	136	128	130	0.00036	0.00059
5	1	124	0.00065	0.00108	137	128	133	0.01595	0.02649
6	1	126	0.00138	0.0023	138	130	131	0.00102	0.00169
7	1	142	0.00193	0.00321	139	130	132	0.00025	0.00042
8	1	155	0.02743	0.04555	140	133	134	0.00005	0.00008
9	1	169	0.04841	0.08039	141	133	135	0.00254	0.00422
10	1	208	0.03982	0.06613	142	135	136	0.01224	0.02033
11	1	225	0.33227	0.55181	143	136	137	0.0093	0.01544
12	1	234	0.00223	0.00371	144	136	138	0.01102	0.0183
13	1	245	0.01077	0.01788	145	138	139	0.00051	0.00084
14	2	3	0.01234	0.0205	146	138	141	0.00195	0.00324
15	2	7	0.05698	0.09462	147	139	140	0.0065	0.0108
16	3	4	0.01625	0.02699	148	142	143	0.06197	0.10291
17	3	5	0.00406	0.00675	149	142	154	0.00508	0.00844
18	3	6	0.01082	0.01797	150	143	144	0.12424	0.20633
19	7	8	0.0093	0.01544	151	144	145	0.00223	0.00371
20	7	10	0.0033	0.00548	152	145	146	0.00104	0.00172
21	7	11	0.00497	0.00825	153	146	147	0.00064	0.00106
22	7	19	0.00547	0.00908	154	147	148	0.00112	0.00186
23	8	9	0.00053	0.00089	155	147	149	0.00091	0.00152
24	11	12	0.00503	0.00835	156	147	150	0.00046	0.00076
25	11	13	0.00924	0.01535	157	147	151	0.00041	0.00067
26	13	14	0.00853	0.01417	158	147	152	0.00066	0.0011

Line index	From bus	To bus	r (Ω)	x (Ω)	Line index	From bus	To bus	r (Ω)	x (Ω)
27	13	16	0.00335	0.00557	159	147	153	0.00914	0.01518
28	13	18	0.00046	0.00076	160	155	156	0.00051	0.00084
29	14	15	0.00091	0.00152	161	156	157	0.00965	0.01603
30	16	17	0.0005	0.00083	162	157	158	0.00635	0.01054
31	19	20	0.00061	0.00101	163	157	159	0.0029	0.00481
32	20	21	0.00005	0.00008	164	159	160	0.00462	0.00768
33	21	22	0.00009	0.00015	165	160	161	0.00036	0.00059
34	21	23	0.00254	0.00422	166	160	163	0.01107	0.01839
35	22	25	0.02941	0.04884	167	160	165	0.01402	0.02328
36	23	24	0.00036	0.00059	168	161	162	0.00254	0.00422
37	26	27	0.01244	0.02067	169	163	164	0.0004	0.00066
38	27	28	0.06131	0.10181	170	165	166	0.00051	0.00084
39	28	29	0.06141	0.10198	171	165	167	0.01117	0.01856
40	29	30	0.04231	0.07027	172	167	168	0.00112	0.00186
41	30	31	0.00071	0.00118	173	169	170	0.06095	0.10122
42	30	32	0.00046	0.00076	174	170	171	0.00569	0.00945
43	30	33	0.01636	0.02716	175	170	175	0.00894	0.01485
44	30	41	0.00274	0.00456	176	171	172	0.00051	0.00084
45	33	34	0.00716	0.01189	177	172	173	0.00051	0.00084
46	33	35	0.00036	0.00059	178	172	174	0.00051	0.00084
47	33	37	0.00091	0.00151	179	175	176	0.00462	0.00768
48	35	36	0.0002	0.00034	180	175	179	0.03774	0.06267
49	37	38	0.00696	0.01156	181	176	177	0.00574	0.00953
50	37	39	0.00051	0.00084	182	177	178	0.00076	0.00127
51	39	40	0.00005	0.00008	183	179	180	0.01559	0.0259
52	41	42	0.00061	0.00101	184	179	181	0.0223	0.03703
53	41	43	0.00076	0.00127	185	179	203	0.00381	0.00633
54	43	44	0.00273	0.00454	186	181	182	0.00005	0.00008
55	43	45	0.00036	0.00059	187	181	189	0.00056	0.00093
56	43	46	0.00554	0.00919	188	182	183	0.00061	0.00101
57	46	47	0.00046	0.00076	189	182	184	0.02184	0.03627
58	46	49	0.01346	0.02235	190	184	185	0.00879	0.01459
59	47	48	0.00081	0.00135	191	184	187	0.00036	0.00059
60	49	50	0.00025	0.00042	192	184	188	0.00538	0.00894
61	50	51	0.00046	0.00076	193	185	186	0.00046	0.00076
62	50	52	0.0129	0.02143	194	189	190	0.01986	0.03298
63	52	53	0.00635	0.01054	195	189	191	0.00792	0.01316
64	52	54	0.00051	0.00084	196	189	192	0.00508	0.00844
65	52	55	0.00036	0.00059	197	192	193	0.02865	0.04758
66	55	56	0.0064	0.01063	198	192	195	0.00051	0.00084
67	56	57	0.00041	0.00067	199	192	196	0.00239	0.00396
68	58	59	0.00315	0.00523	200	193	194	0.0029	0.00481
69	58	60	0.00046	0.00077	201	196	197	0.00137	0.00228

Line index	From bus	To bus	r (Ω)	x (Ω)	Line index	From bus	To bus	r (Ω)	x (Ω)
70	58	62	0.01966	0.03264	202	196	198	0.00213	0.00354
71	60	61	0.01514	0.02514	203	198	199	0.01366	0.02269
72	62	63	0.00036	0.00059	204	198	200	0.01717	0.02851
73	63	64	0.00437	0.00725	205	200	201	0.00559	0.00928
74	63	65	0.0003	0.00051	206	200	202	0.00046	0.00076
75	63	76	0.00513	0.00852	207	203	204	0.00051	0.00084
76	65	66	0.01189	0.01974	208	203	205	0.00046	0.00076
77	65	67	0.02072	0.03442	209	203	206	0.00051	0.00084
78	65	68	0.01666	0.02767	210	203	207	0.0002	0.00034
79	68	69	0.00188	0.00312	211	208	209	0.01051	0.01746
80	68	70	0.01173	0.01949	212	208	210	0.00655	0.01088
81	68	72	0.00178	0.00295	213	209	224	0.0003	0.00051
82	70	71	0.00904	0.01501	214	210	211	0.01214	0.02016
83	72	73	0.00025	0.00042	215	210	214	0.00117	0.00194
84	73	74	0.01265	0.021	216	211	212	0.00062	0.00102
85	73	75	0.04663	0.07744	217	212	213	0.00048	0.00079
86	76	77	0.00137	0.00228	218	214	215	0.00234	0.00388
87	78	79	0.02225	0.03695	219	214	217	0.00726	0.01206
88	79	80	0.08833	0.14669	220	215	216	0.0003	0.00051
89	80	81	0.02067	0.03433	221	217	218	0.00025	0.00042
90	80	82	0.0382	0.06343	222	217	219	0.00173	0.00287
91	80	88	0.01377	0.02286	223	217	220	0.00163	0.0027
92	81	122	0.02133	0.03543	224	217	221	0.00081	0.00135
93	82	83	0.0003	0.00051	225	221	222	0.0253	0.04201
94	82	84	0.00025	0.00042	226	222	223	0.00046	0.00076
95	84	85	0.00701	0.01164	227	225	226	0.00051	0.00084
96	84	86	0.00132	0.00219	228	225	227	0.02022	0.03358
97	84	123	0.0003	0.00051	229	225	228	0.02321	0.03854
98	86	87	0.00041	0.00067	230	225	229	0.02338	0.03883
99	88	89	0.06151	0.10215	231	225	230	0.01833	0.03043
100	89	90	0.00102	0.00169	232	225	231	0.00759	0.0126
101	89	93	0.0158	0.02623	233	231	232	0.01135	0.01885
102	89	95	0.00025	0.00042	234	232	233	0.01272	0.02113
103	91	92	0.00046	0.00076	235	234	235	0.07096	0.11784
104	92	93	0.00041	0.00067	236	235	236	0.07609	0.12636
105	93	94	0.00025	0.00042	237	236	237	0.00665	0.01105
106	95	96	0.02088	0.03467	238	236	244	0.00066	0.0011
107	95	97	0.00919	0.01527	239	237	238	0.00233	0.00387
108	95	98	0.0003	0.00051	240	238	239	0.00183	0.00304
109	98	99	0.02138	0.03551	241	239	240	0.0003	0.00051
110	99	100	0.01813	0.03011	242	239	241	0.00274	0.00456
111	99	120	0.00706	0.01173	243	239	242	0.00305	0.00506
112	100	101	0.0002	0.00034	244	242	243	0.00071	0.00118

Line index	From bus	To bus	r (Ω)	x (Ω)	Line index	From bus	To bus	r (Ω)	x (Ω)
113	100	115	0.02606	0.04327	245	245	246	0.0189	0.03138
114	100	116	0.02352	0.03906	246	245	257	0.00208	0.00346
115	101	102	0.00036	0.00059	247	246	251	0.0003	0.00051
116	101	109	0.00483	0.00801	248	246	247	0.00914	0.01518
117	102	103	0.0029	0.00481	249	247	248	0.00036	0.00059
118	103	104	0.00036	0.00059	250	248	249	0.00015	0.00025
119	104	105	0.0002	0.00034	251	249	250	0.00046	0.00076
120	104	108	0.00041	0.00067	252	251	252	0.00117	0.00194
121	105	106	0.00056	0.00093	253	251	265	0.00254	0.00422
122	105	107	0.00056	0.00093	254	252	253	0.00122	0.00202
123	109	110	0.03571	0.0593	255	252	254	0.0002	0.00034
124	109	112	0.01148	0.01906	256	254	255	0.0002	0.00034
125	110	111	0.00041	0.00067	257	255	256	0.00057	0.00094
126	112	113	0.00071	0.00118	258	257	258	0.00254	0.00422
127	113	114	0.00076	0.00127	259	257	259	0.00747	0.0124
128	116	117	0.00477	0.00793	260	257	262	0.00467	0.00776
129	116	118	0.00173	0.00287	261	259	260	0.00046	0.00076
130	118	119	0.00036	0.00059	262	259	261	0.0002	0.00034
131	120	121	0.00508	0.00844	263	262	263	0.00325	0.0054
132	124	125	0.00475	0.00789	264	262	264	0.00102	0.00169

C. Microgrids configurations and parameters

The microgrids are indexed by k and assigned to two types: wind-based Type A for odd k and photovoltaic-based Type B for even k. The two types differ in their renewable generation sources, storage capacities, and load profiles. Within each type, the quadratic and linear coefficients of the generator cost functions vary with k, introducing additional heterogeneity as the two configurations are replicated to increase the number of microgrids.

TABLE VIII
ENERGY STORAGE PARAMETERS

MG type and index k	$P_i^{E,max}$ (kW)	$E_{i,max}$ (kWh)	$E_{i,min}$ (kWh)	b_i^E ($/kWh)	c_i^E ($)	η_i
Type A, k=1, 3, 5, …	30	300	0	1.389e-6	0.0764	0.95
Type B, k=2, 4, 6, …	20	200	0	1.944e-6	0.0903	0.95

TABLE IX
GENERATOR PARAMETERS

MG type and index k	$P_i^{G,max}$ (kW)	$P_i^{G,min}$ (kW)	R_i^G (kW/h)	a_i^G ($/kW²h)	b_i^G ($/kWh)	c_i^G ($)
Type A, k=1, 3, 5, …	500	0	500	4.3e-5 + 5.0e-8 × k	0.033 + 5.0e-5 × k	0.435
Type B, k=2, 4, 6, …	500	0	500	3.7e-5 + 5.0e-8 × k	0.034 + 5.0e-5 × k	0.525